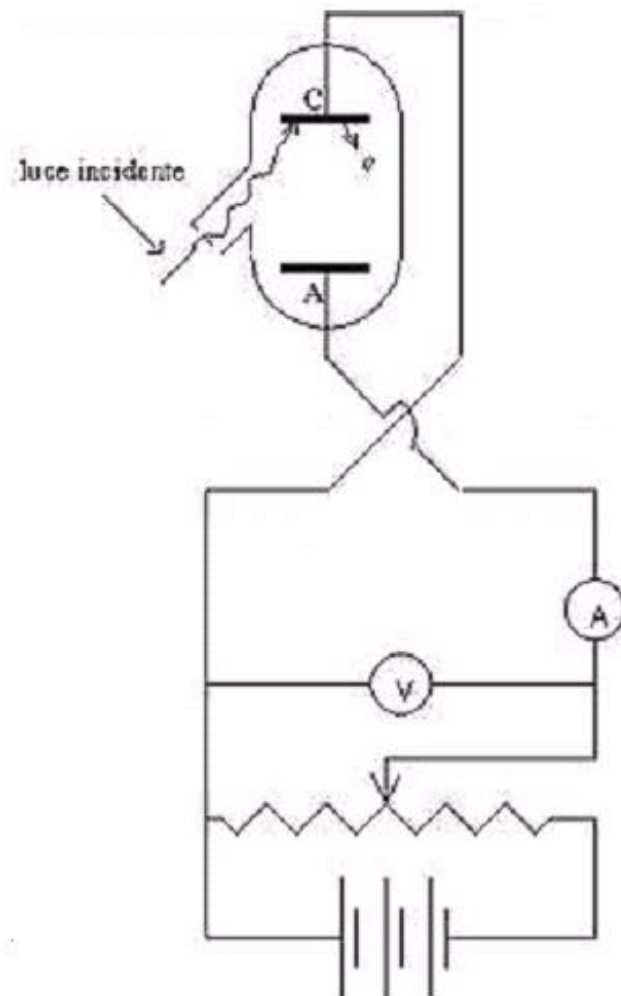
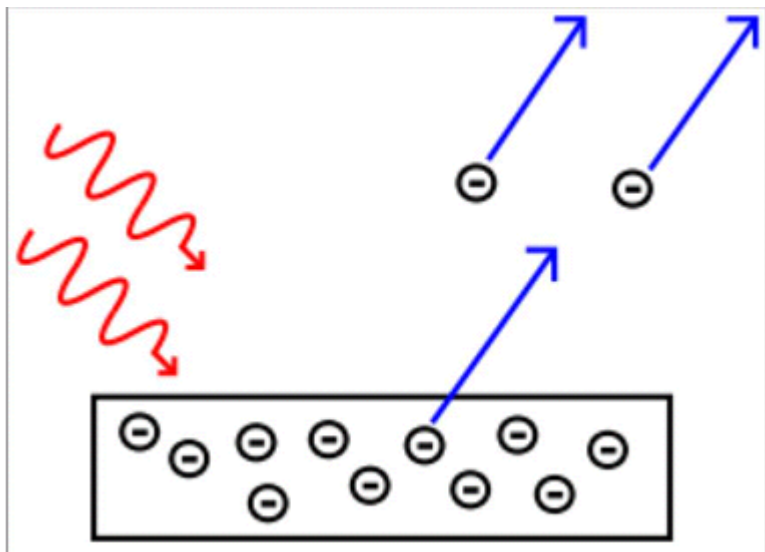
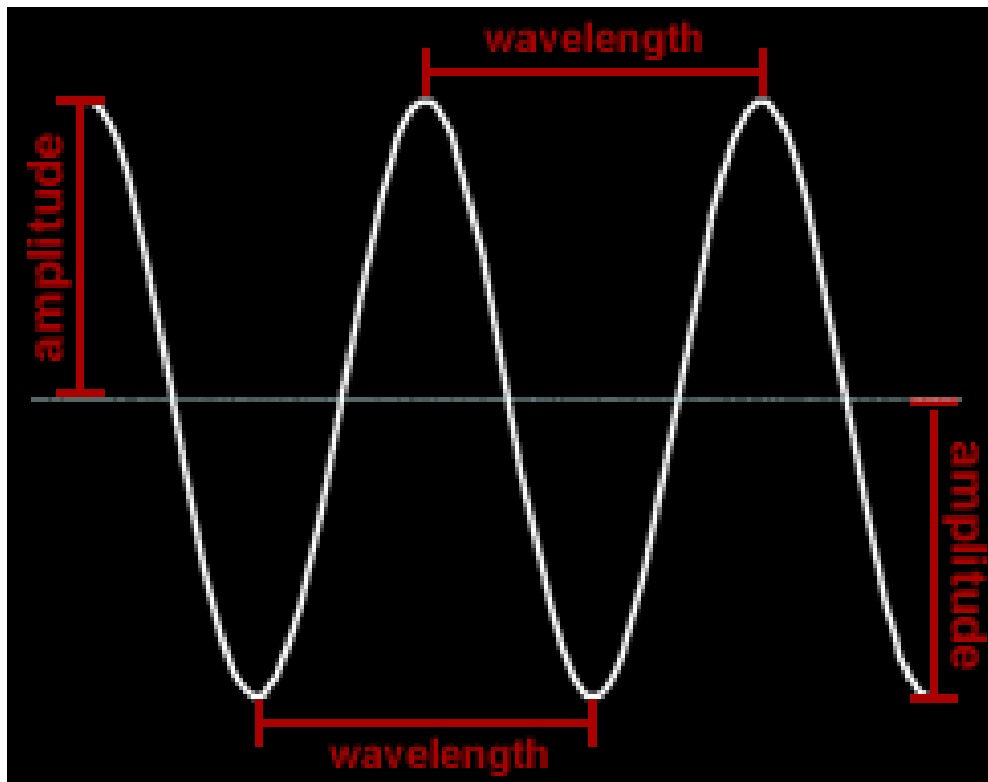


Effetto Fotoelettrico



Ottica ondulatoria

Fino al 1900, la teoria ondulatoria della luce (elettromagnetismo classico) descriveva la maggior parte dei fenomeni noti.



Onde elettromagnetiche:

caratterizzate da:

- Ampiezza (A)
- Frequenza (ν)
- Lungh. d'onda (λ)

Flusso di Energia $\propto A^2$

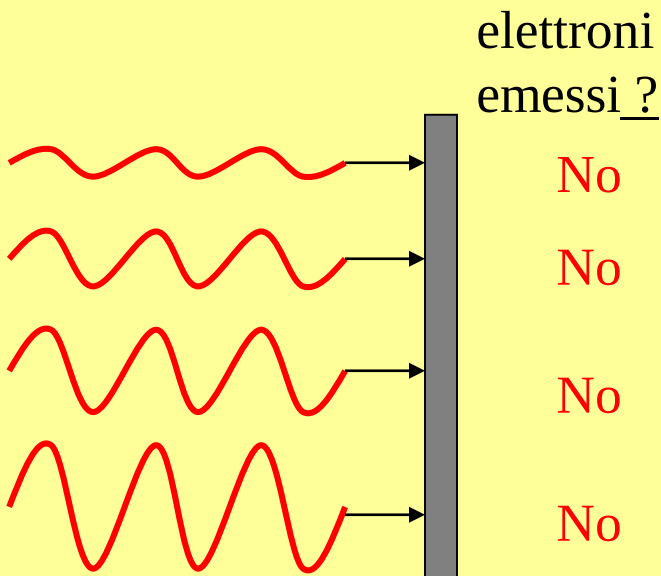
Effetto Fotoelettrico (I)

Attese “Classiche”

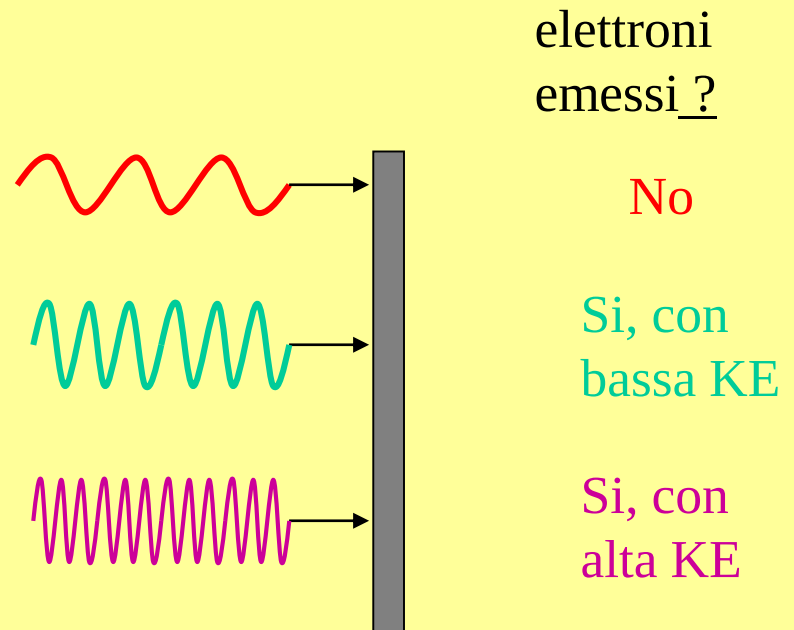
Incremento dell'energia cinetica con l'ampiezza dell'onda:
emissione al di sopra di una INTENSITÀ di soglia

Invece:

Variando l'AMPIEZZA a frequenza costante

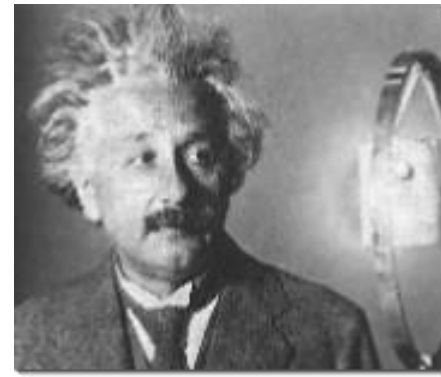


Variando la FREQUENZA ad ampiezza costante



- emissione solo se $\nu > \nu_0$ (dipendente dal materiale;)
- corrente elettronica PROPORZIONALE all'Intensità della radiazione

Effetto Fotoelettrico (II)

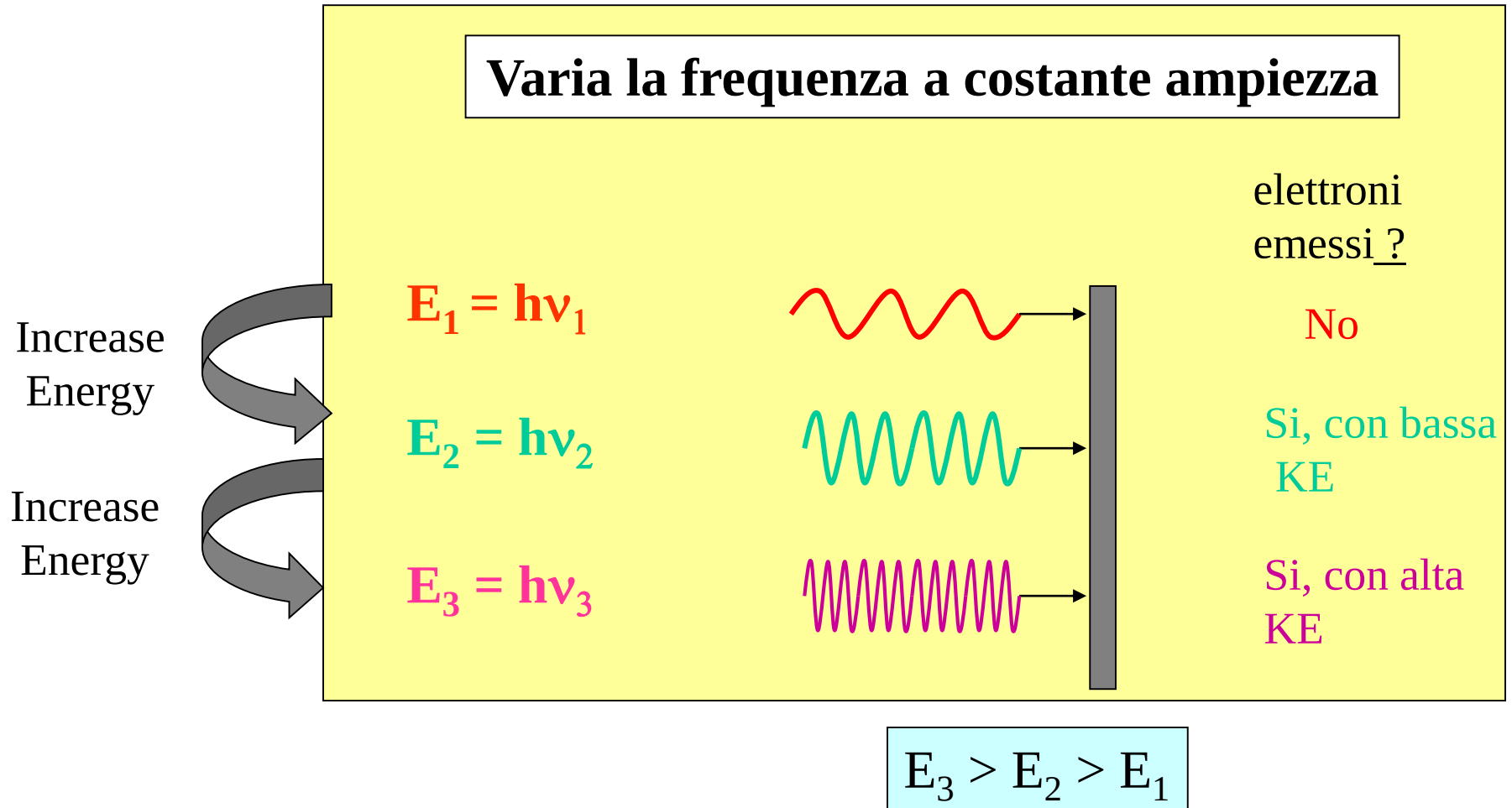


❑ la spiegazione è che il campo di radiazione scambia energia con la materia secondo quantità finite, dette *quanti* : **$E=h\nu$**

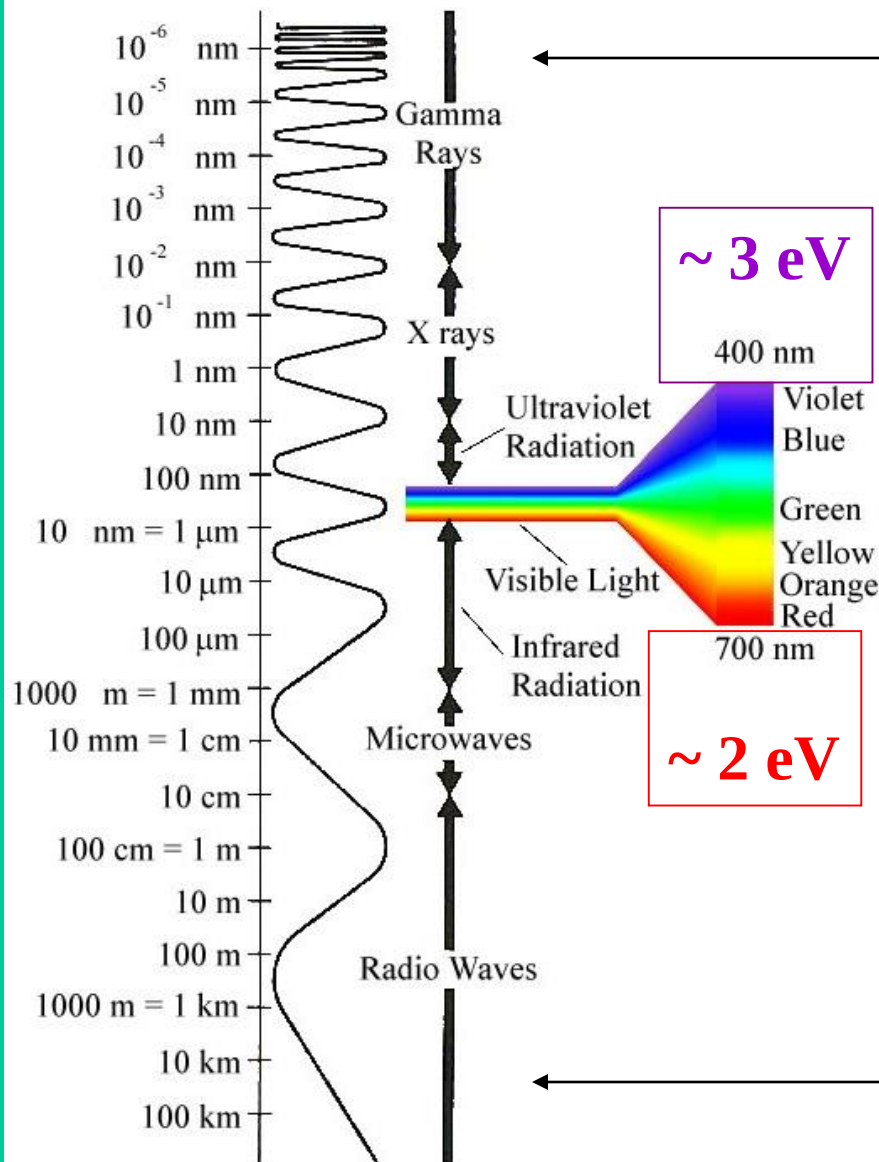
❑ **l'energia dei quanti d'energia** è **proporzionale alla frequenza**: la **soglia fotoelettrica** corrisponde quindi ad una soglia in **FREQUENZA**

❑ **aumentare l'Ampiezza significa aumentare il numero di quanti, e NON aumentare l'energia di ognuno !**

Interpretazione dell' Effetto Fotoelettrico



Lo Spettro elettromagnetico



Lunghezze d'onda minori
(fotoni più energetici)

$$E = h\nu = hc/\lambda$$

$$h = 4.136 \times 10^{-15} [\text{eV} \cdot \text{sec}]$$

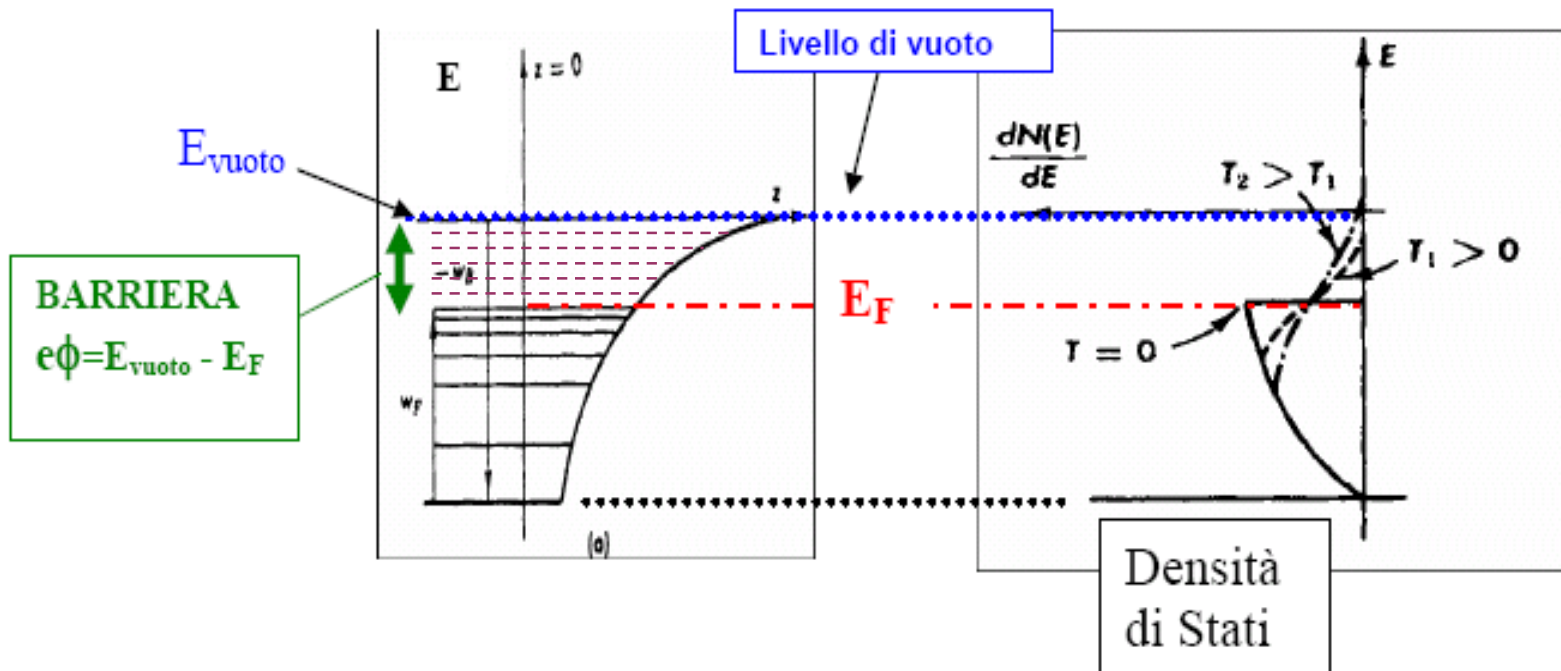
(costante di Planck)

$$E[\text{eV}] = \frac{1240}{\lambda[\text{nm}]}$$

Lunghezze d'onda maggiori
(fotoni meno energetici)

Struttura elettronica (energetica) di un materiale (metallo): Funzione lavoro

- ❑ gli elettroni sono legati al materiale, serve una certa energia per liberarli
- ❑ com'è fatta la struttura elettronica di un generico materiale: caso di un METALLO



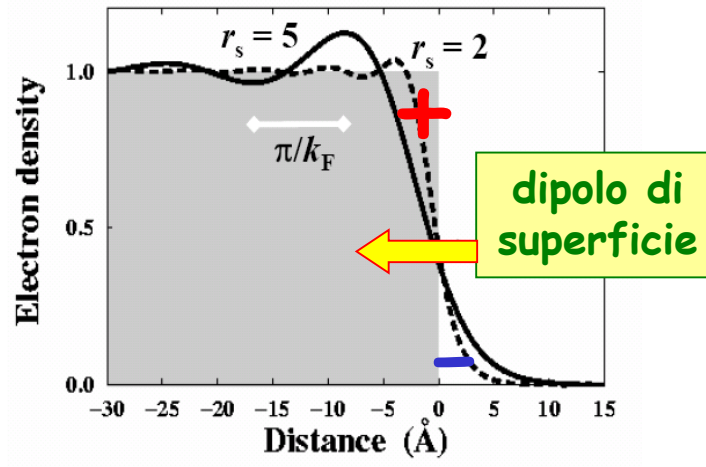
- ❑ Nei metalli, l'energia di legame degli elettroni più esterni (delocalizzati) è la barriera superficiale, detta funzione lavoro (Work Function) $e\Phi$

Barriera Superficiale (Funzione lavoro): origine

$\Phi =$

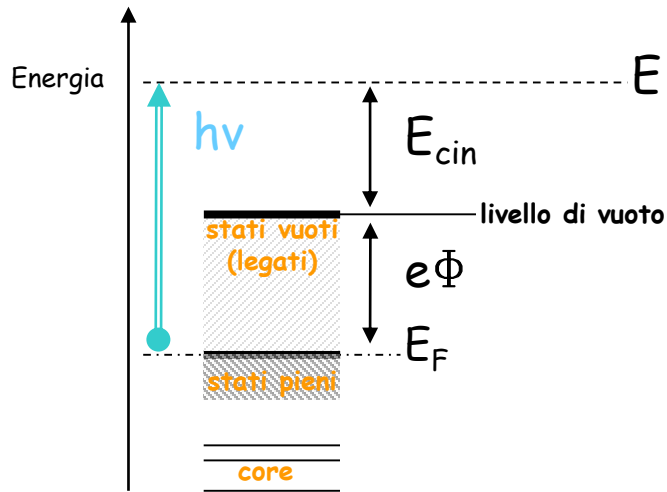
+

carica immagine



Distribuzione in energia della corrente Fotoelettrica

diagramma energetico



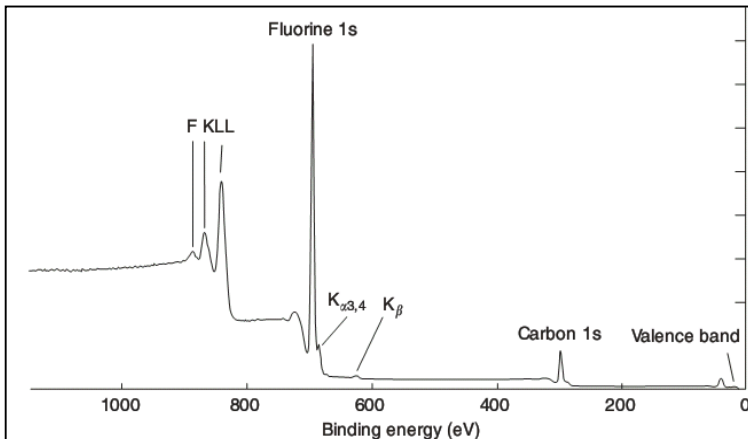
(massima) energia cinetica dell'elettrone fotoemesso

$$E_{cin}^{max} = h\nu - e\phi$$

frequenza di soglia

$$E_{cin} > 0 \Rightarrow \nu > \nu_{soglia}$$

$$\nu_{soglia} = \frac{e\phi}{h}$$



Tipico spettro di fotoemissione eccitato da raggi X (XPS)

$$E_{cin} = h\nu - E_B - e\phi$$

Misura di h mediante l'effetto Fotoelettrico

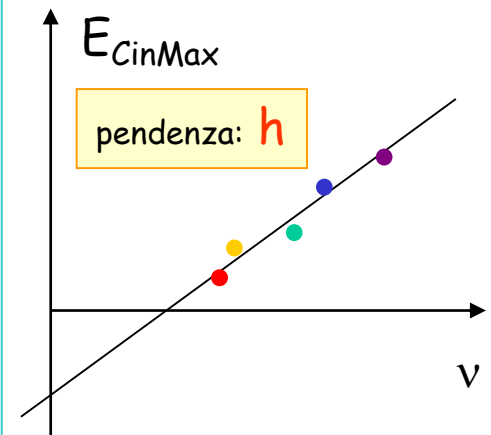
(massima) energia cinetica
dell'elettrone fotoemesso

$$E_{cin}^{\max} = h\nu - e\phi$$

la costante di Planck è il coefficiente angolare di una
retta:

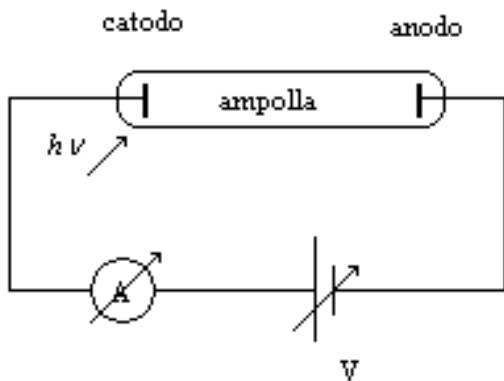
per determinarne il valore ci basta **misurare l'energia
cinetica massima** degli elettroni fotoemessi a diversi
valori della frequenza del fotone

e quindi eseguire una **regressione lineare di E_{CinMax} vs ν** .

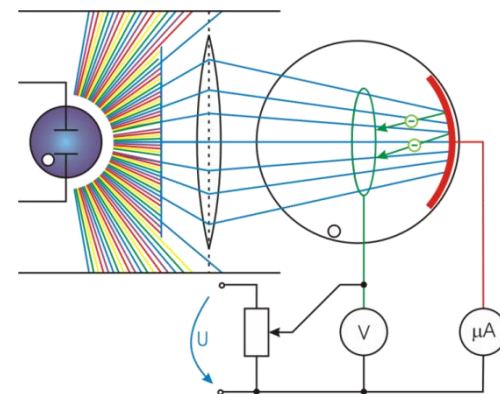
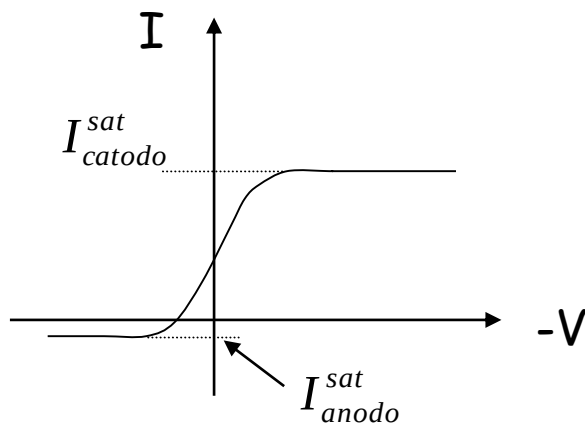


ma come misurare E_{CinMax} ???

MISURA DI E_{CinMax}

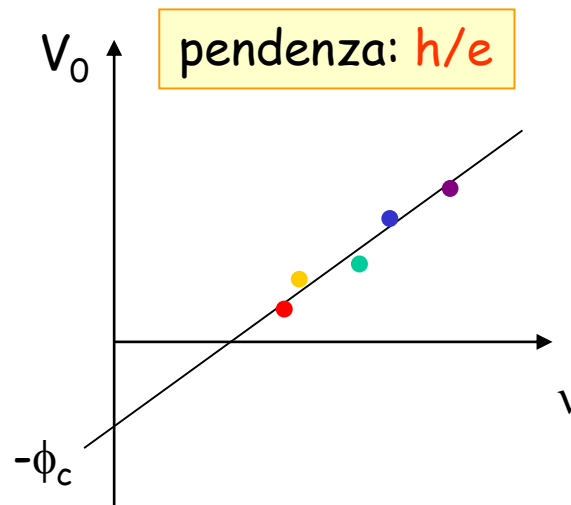


potenziale ritardante V
applicato tra anodo e catodo

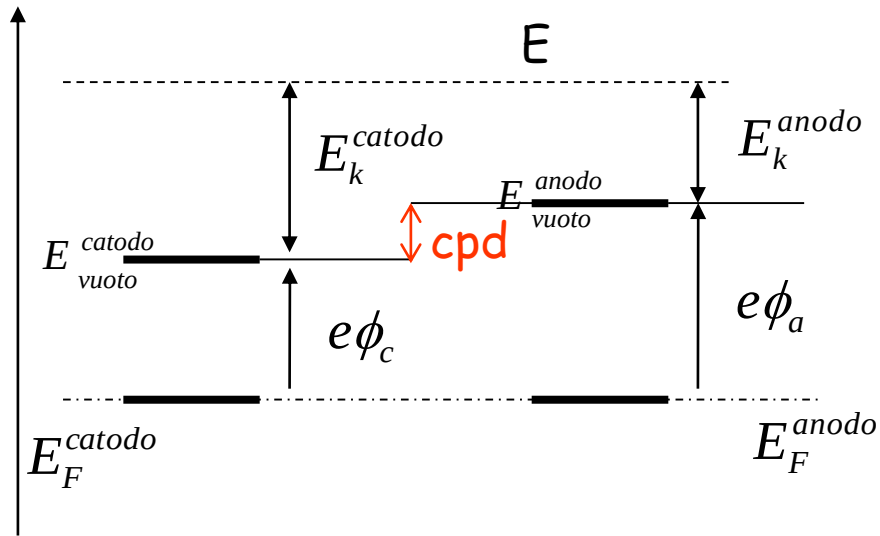


potenziale di arresto $V_0 = E_{\text{CinMax}}/e$

$$eV_0 = h\nu - e\phi_c \Rightarrow V_0(\nu) = -\phi_c + \frac{h}{e}\nu$$

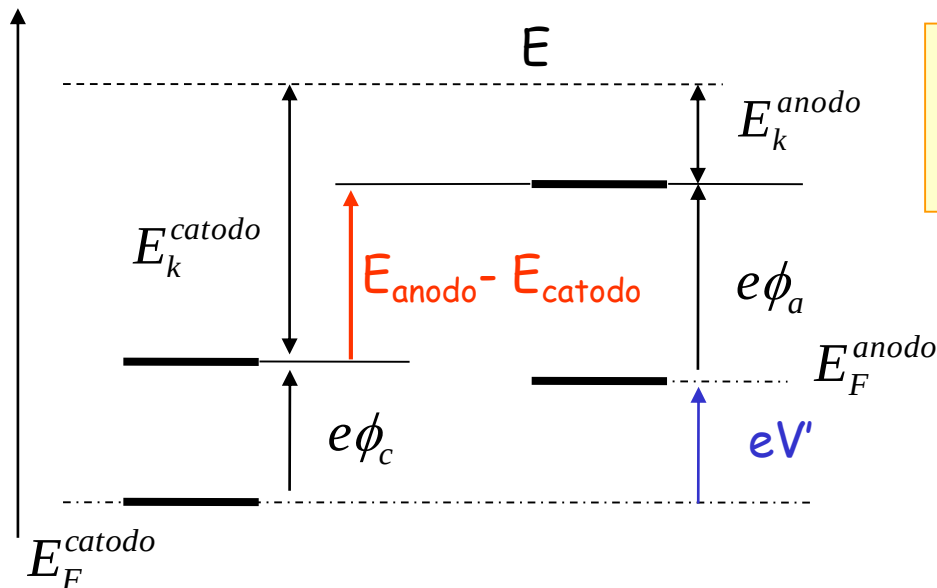
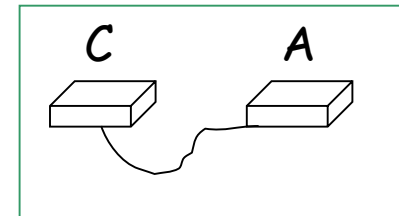


Potenziale applicato(V') vs. potenziale sperimentato dagli elettroni (V)



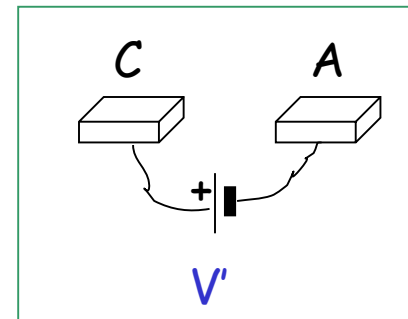
ddp tra anodo e catodo connessi elettricamente tra loro

$$E_{anodo} - E_{catodo} = \phi_a - \phi_c \equiv cpd$$

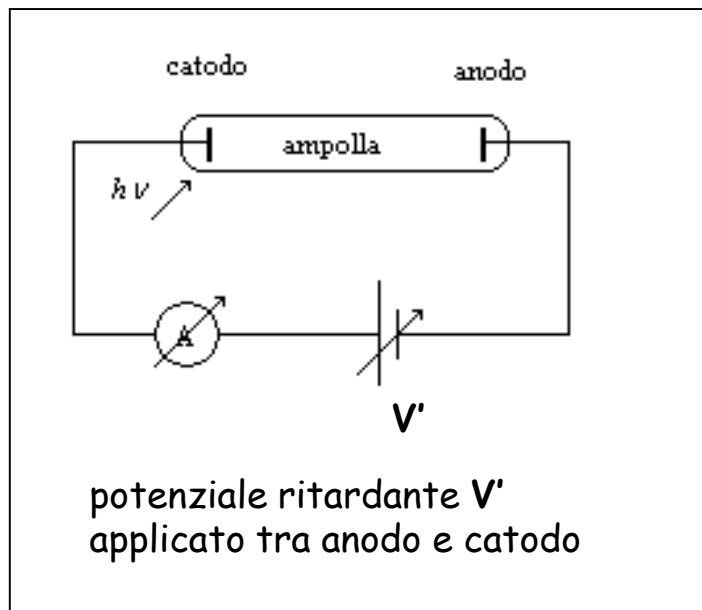


ddp **CORRETTA** tra anodo e catodo

$$E_{anodo} - E_{catodo} = eV' + e\phi_a - e\phi_c$$



Determinazione della costante di Planck



La ddp corretta tra anodo e catodo è allora:

$$eV = E_{\text{anodo}} - E_{\text{catodo}} = eV' + e\phi_a - e\phi_c$$

che specializzata per il potenziale di arresto V_0

$$eV'_0 + e\phi_a - e\phi_c = eV_0 = h\nu - e\phi_c$$

ovvero

$$eV'_0 = h\nu - e\phi_a \Rightarrow V'_0(\nu) = -\phi_a + \frac{h}{e}\nu$$

nella formulazione corretta $\phi_c \rightarrow \phi_a$

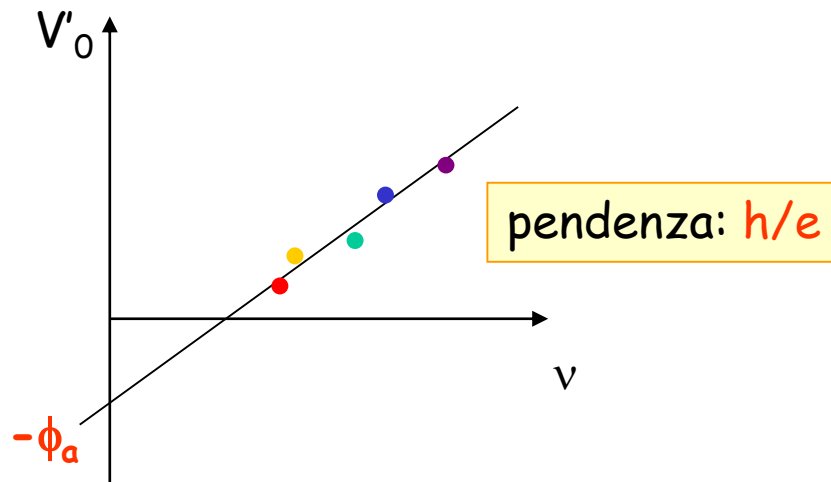
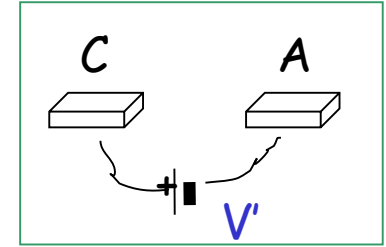
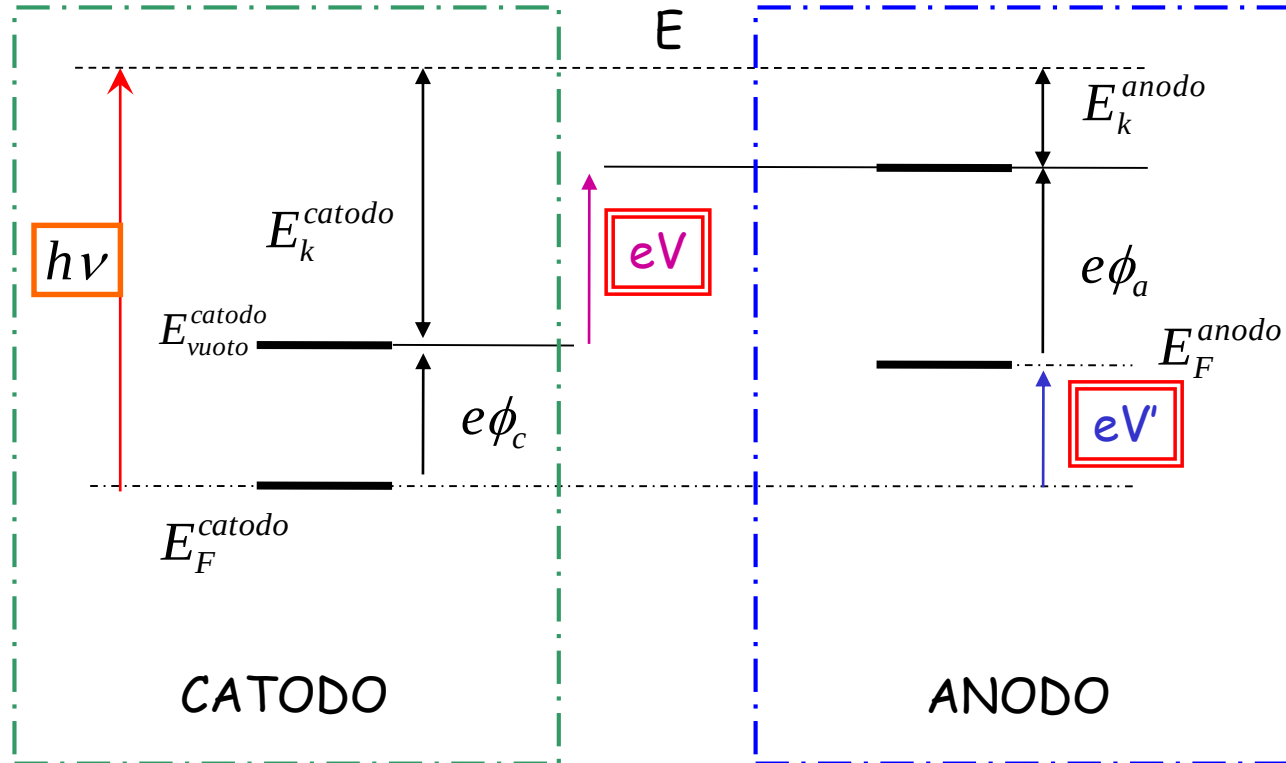


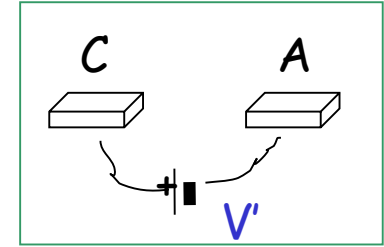
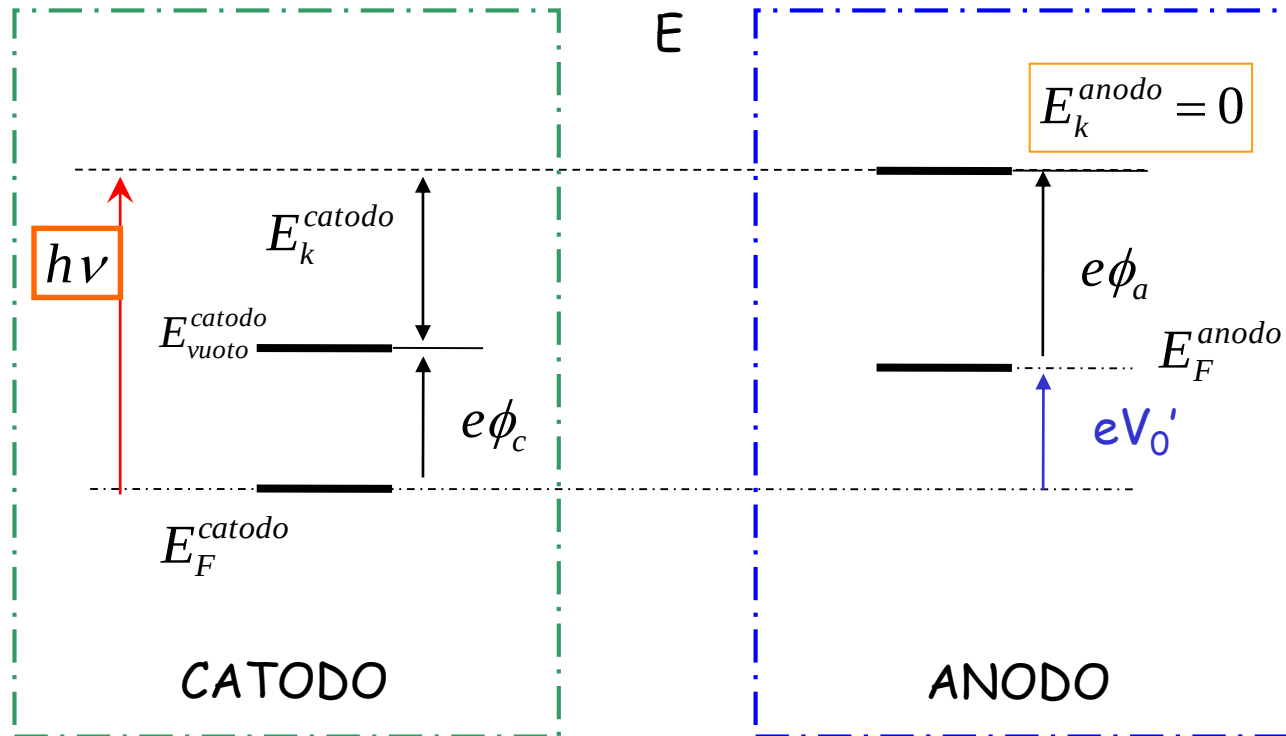
diagramma energetico: eV vs. eV'



per definizione

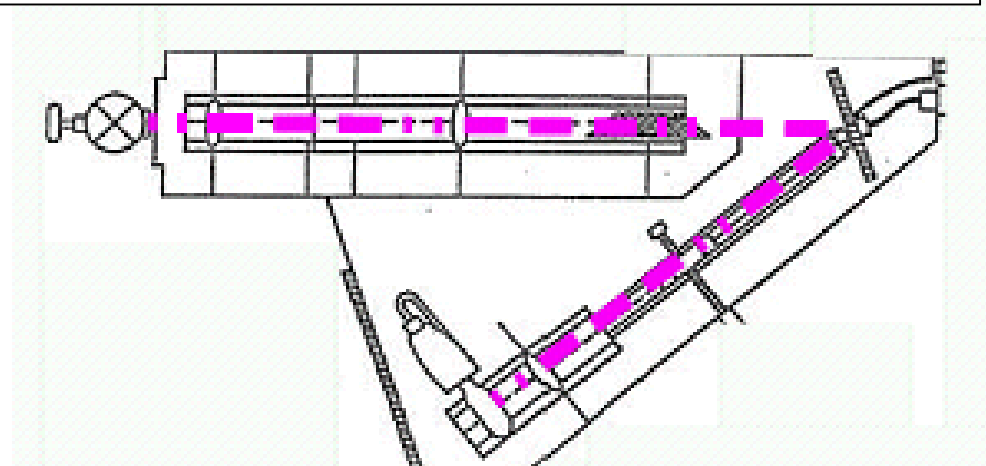
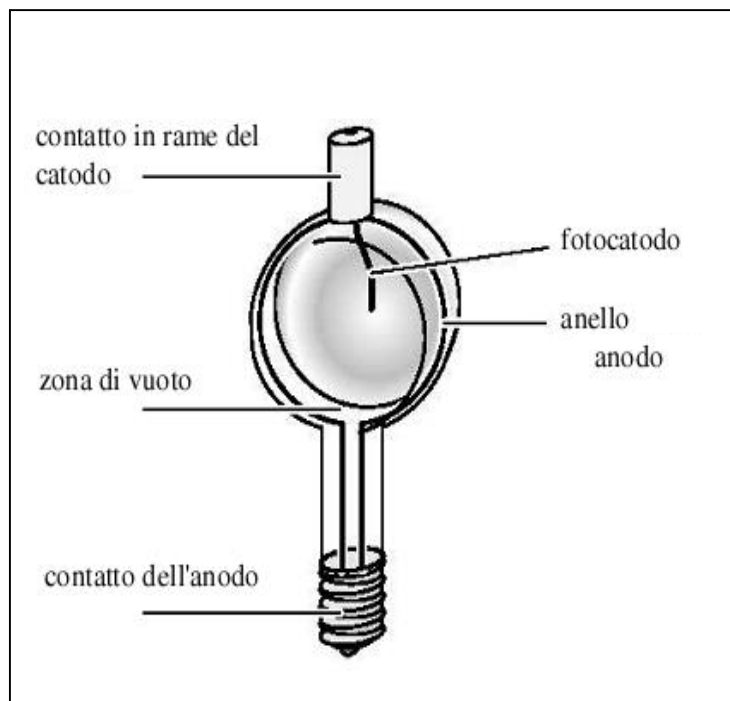
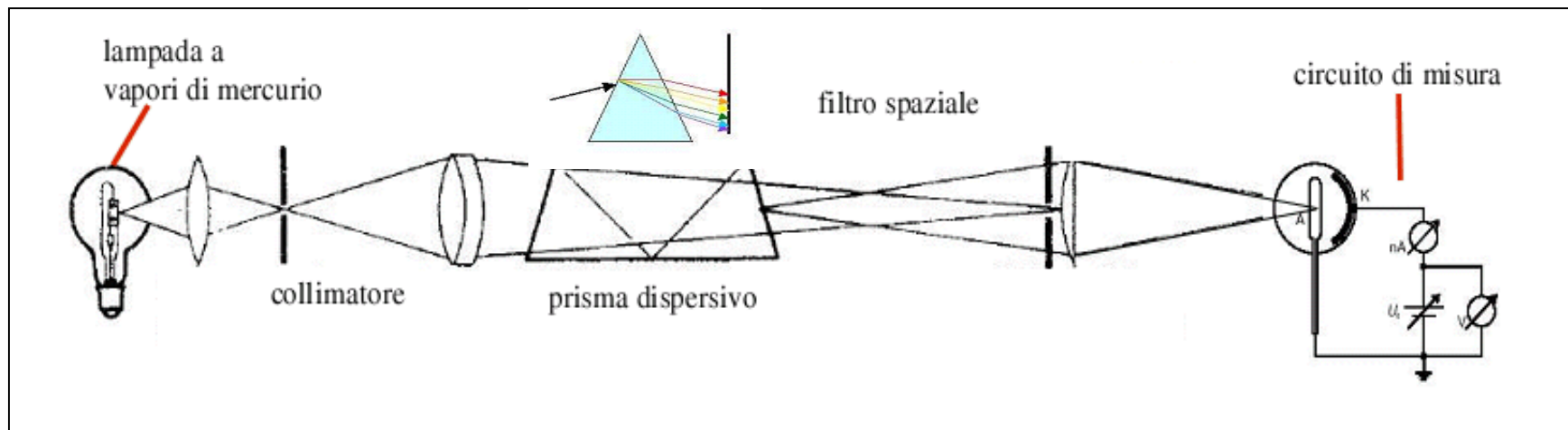
il potenziale di arresto corrisponde a $E_k^{anodo} \equiv 0$

diagramma energetico: potenziale di arresto V_0'

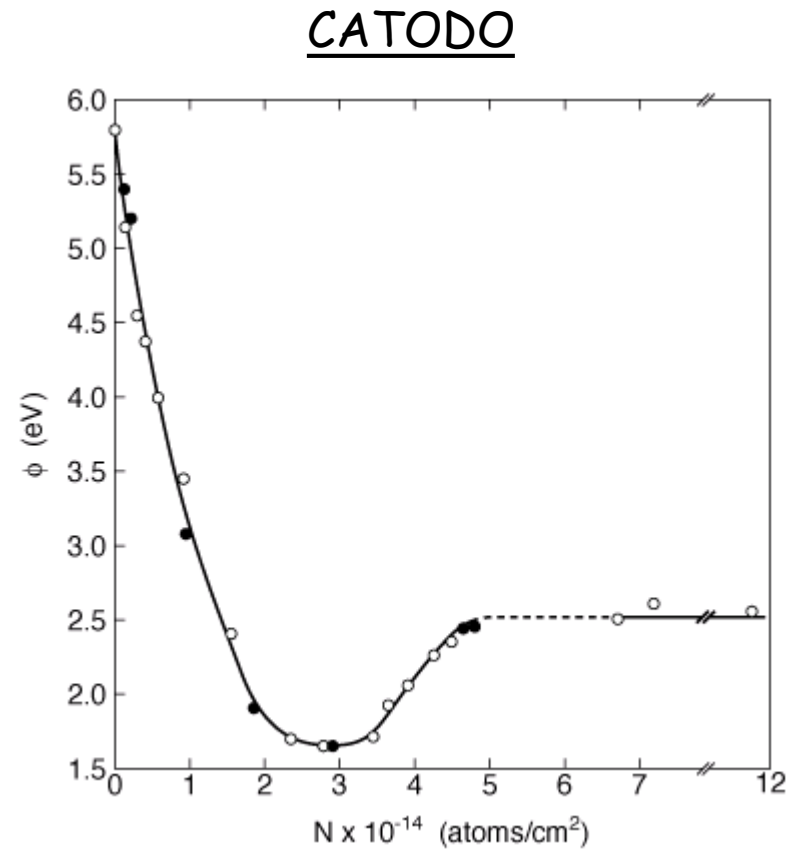
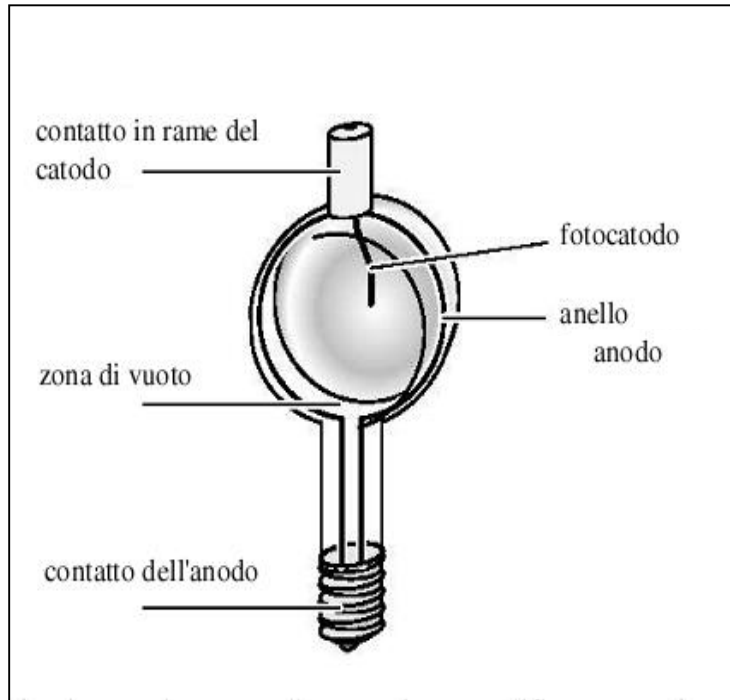


potenziale di arresto $\Leftrightarrow E_k^{anodo} \equiv 0 \Leftrightarrow eV_0' + e\phi_a = h\nu$

$$eV_0' = h\nu - e\phi_a \Rightarrow V_0'(\nu) = -\phi_a + \frac{h}{e}\nu$$



COLORE	FREQUENZA ($\times 10^{14}$ Hz)	LUNGHEZZA D'ONDA (nm)	NOTE
ROSSO	4.53	$\sim 663^4$	la riga e' molto debole
GIALLO	5.19	578	la riga in realta' e' doppia
VERDE	5.49	546	
TURCHESE	6.88	492	molto debole, e non utilizzata
BLU	6.98	436	anche altre due meno significative
VIOLETTO	7.41	405	



*Funzione Lavoro di K/W(110) in
funzione della quantità di potassio (K).*

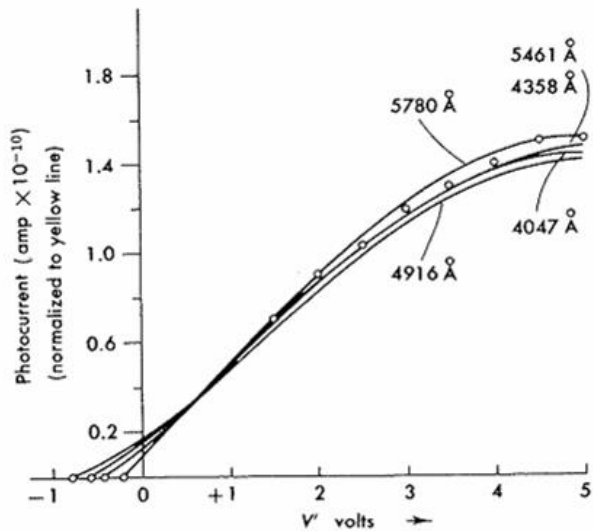


FIG. 1.17 Photocurrent as a function of anode voltage. The currents have been normalized to the yield from the yellow lines ($\lambda = 5780 \text{ \AA}$).

Curve «normalizzate»

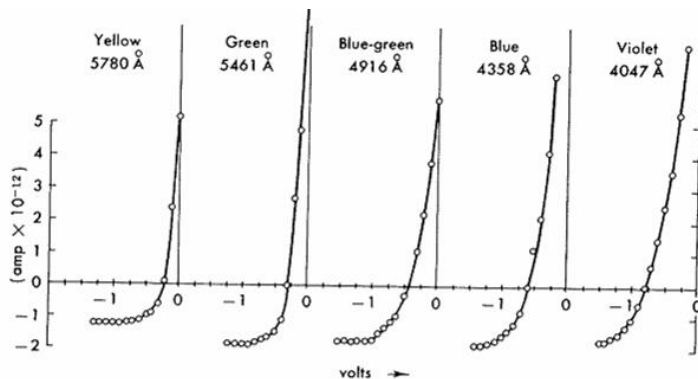
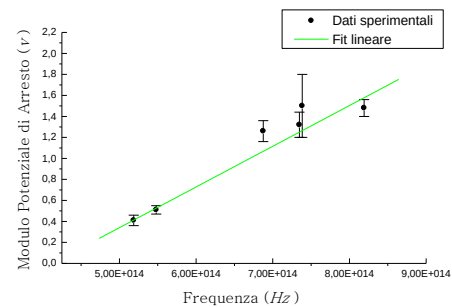
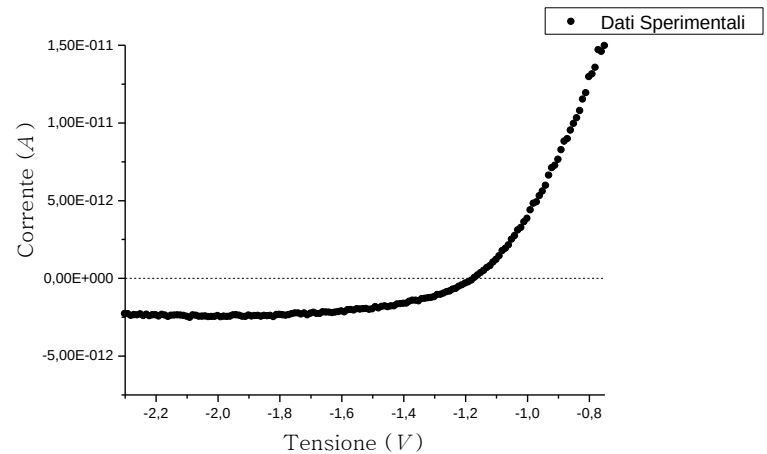
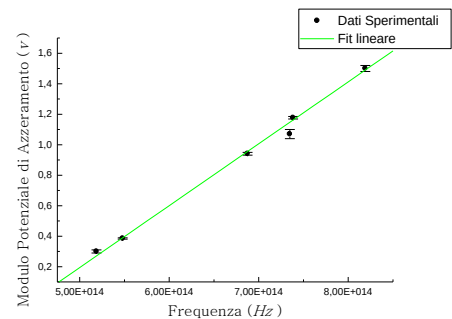


FIG. 1.18 Photocurrent for different wavelengths as a function of stopping potential; the same normalization as in Fig. 1.17 has been used. Note the reverse current due to photoemission from the anode.



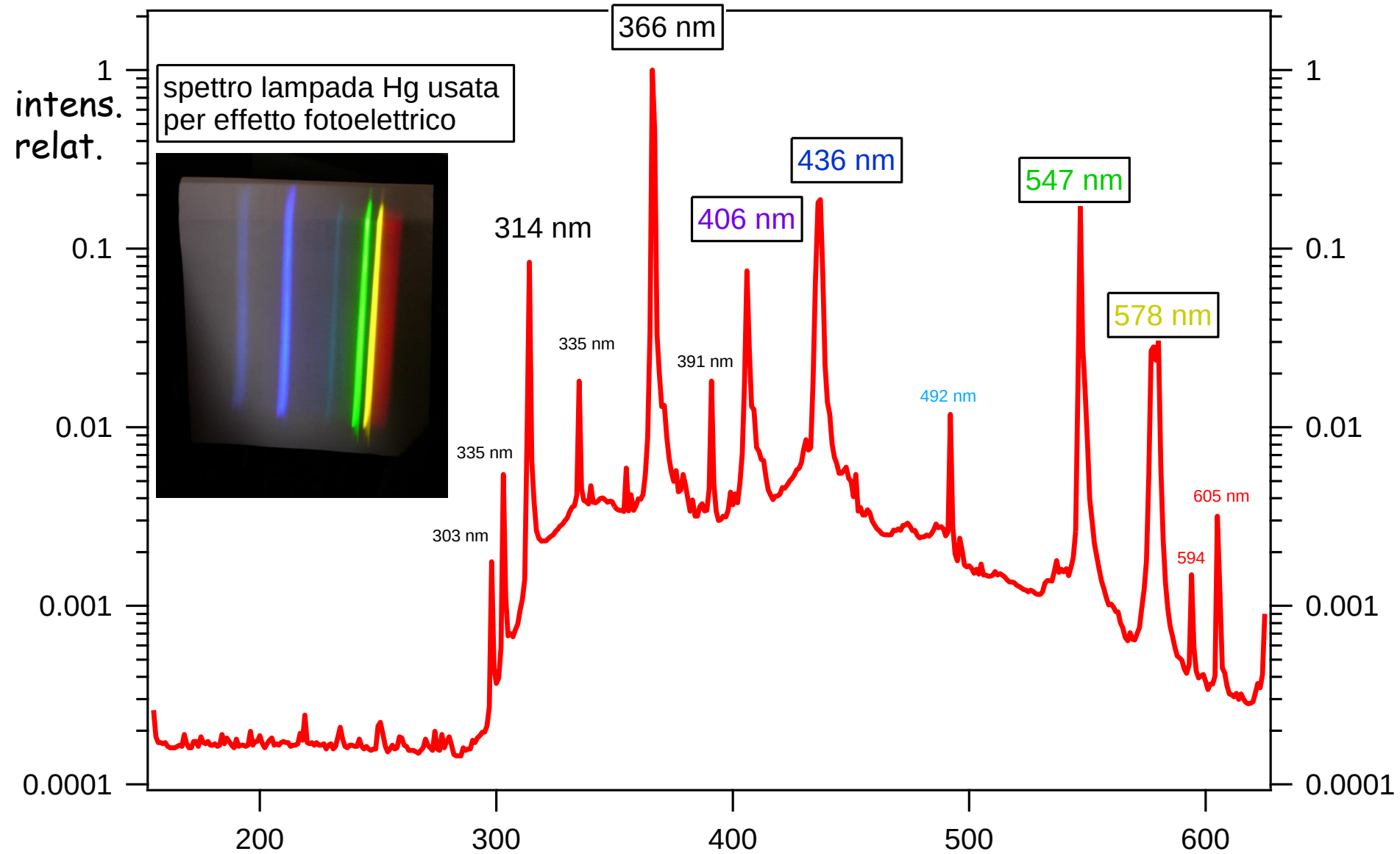
$$h = (6,2 \pm 0,5) \times 10^{-34} \text{ J} \cdot \text{s}$$

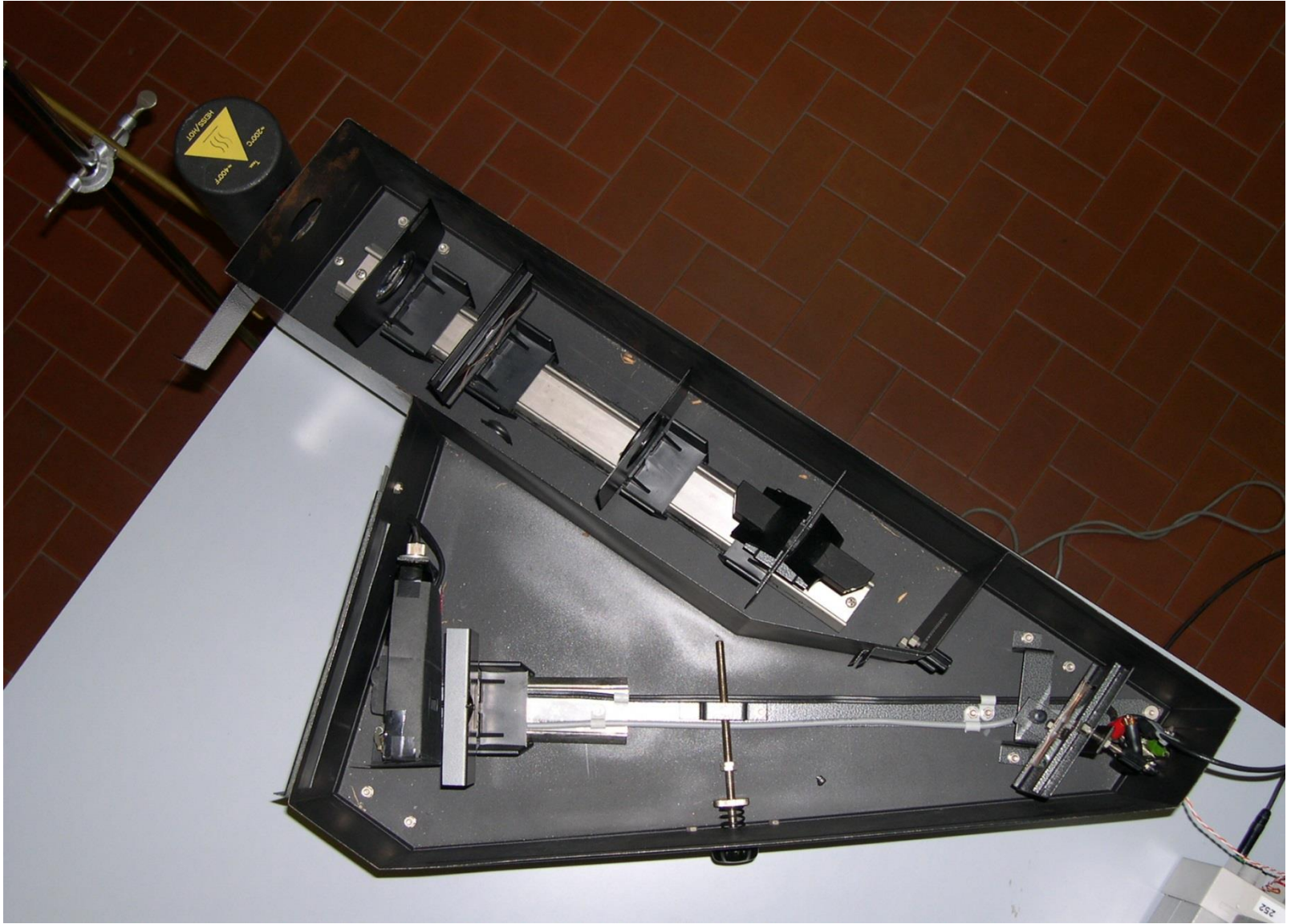
$$h = 6,626 \times 10^{-34} \text{ J} \cdot \text{s}$$



$$h = (6,50 \pm 0,14) \times 10^{-34} \text{ J} \cdot \text{s}$$

Spettro di vapori di Mercurio (bassa pressione)

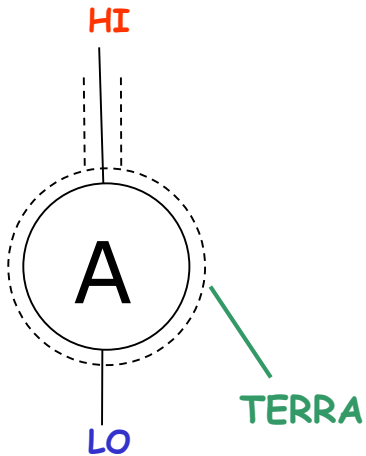




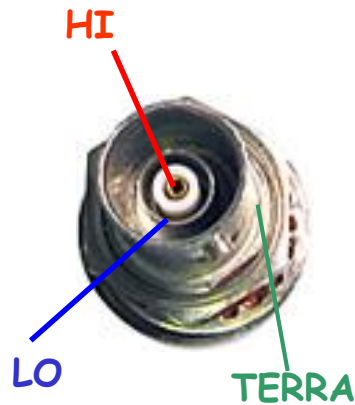
PicoAmperometro Keithley 6487 con sorgente di tensione

Function	Reading Range
Amps	$\pm 10\text{fA}$ to $\pm 21\text{mA}$

Function	Output Range	Available Ranges
Source voltage	$\pm 200\mu\text{V}$ to $\pm 505\text{V}$	10V, 50V, 500V



connettore TRIAX



Procedure

Perform the following steps to measure current:

Step 1. Select current function

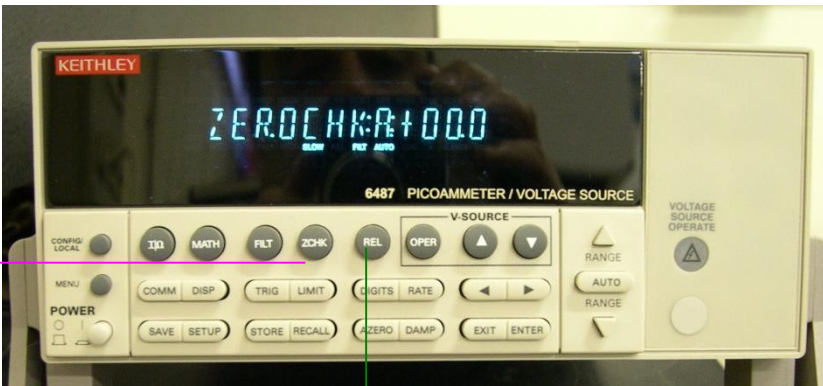
For the Model 6487, press the **IL Ω** key to make sure the current function is selected.

Step 2. Enable zero check

Zero check should always be enabled before making connection changes. The **ZCHK** key toggles zero check on and off.

Step 3. Perform zero correction

To achieve optimum accuracy for low current measurements, it is recommended that you zero correct the picoammeter. To do so, make sure the 2nA range is selected, then press the **ZCOR** key (Model 6485) or **REL** key (Model 6487) to perform zero correction ("ZZ" indicated for Model 6485; MON indicator on for Model 6487).



Step 4. Select a manual measurement range or enable auto range

Use the manual **RANGE** keys to select a manual measurement range or press **AUTO** to enable auto range. With auto range enabled, the instrument will automatically go to the most sensitive range to make the measurement. See [Section 4](#) for details on range.

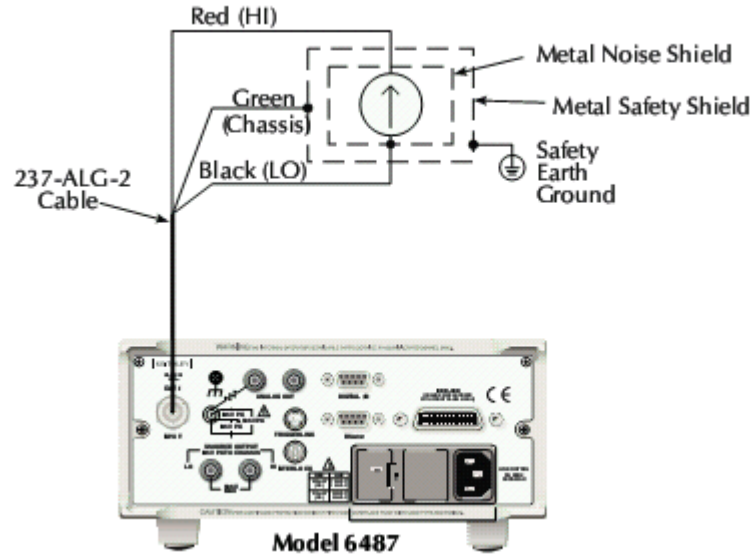
Step 5. Connect the current to be measured to the picoammeter

Basic connections for measurements are shown in [Figure 3-1](#) and [Figure 3-2](#).

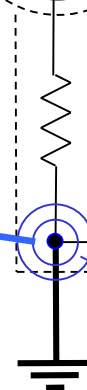
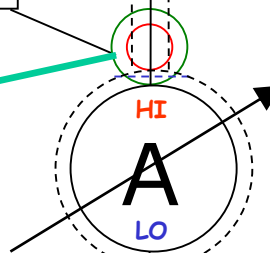
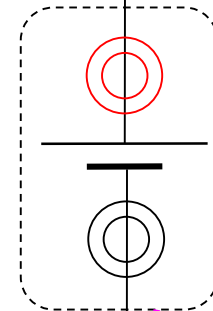
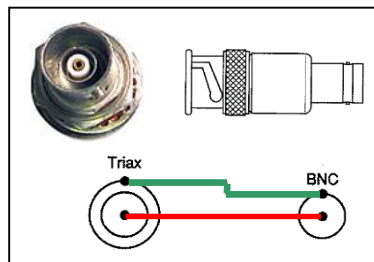
WARNING A safety shield is advisable whenever floating measurements are being made. Connections for the safety shield are shown in [Figure 3-1](#) and [Figure 3-2](#). The metal safety shield must completely surround the noise shield or floating test circuit, and it must be connected to safety earth ground using #18 AWG or larger wire.

NOTE When not making floating measurements, it is recommended that you ground measurement **LO** at only one place in the circuit, such as with the ground link connection on the rear panel of the 6485/6487. (See ["Ground loops," page B-2.](#))
Fundamental information on making connections to the picoammeter input is provided in [Section 2](#).

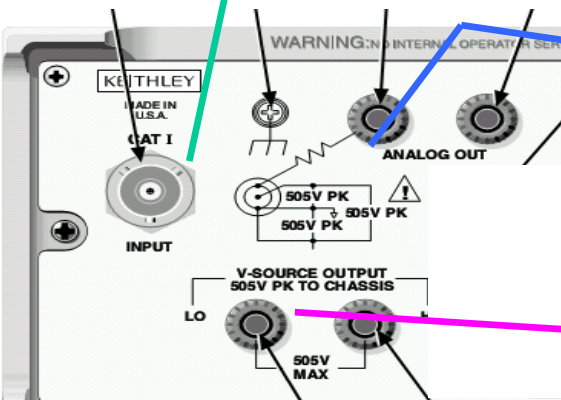
Connections for Model 6487 current measurements



Connessioni



ANALOG OUT NERO

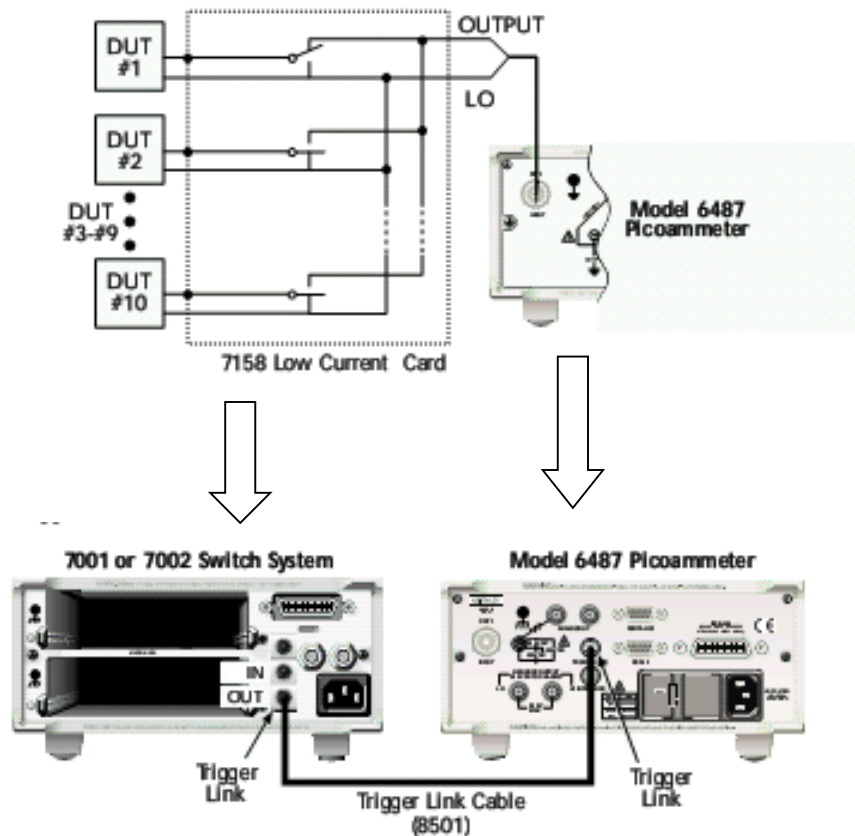


TRIGGERING: quando serve sincronismo con apparati esterni

External trigger example

In a simple test system, you may want to close a switching channel and measure the current from a DUT connected to that channel. Such a test system is shown in [Figure 7-7](#). This example uses a Model 6487 to measure 10 DUTs switched by a Model 7158 low current card in a Model 7001 or 7002 switch system.

Figure 7-7
DUT test system



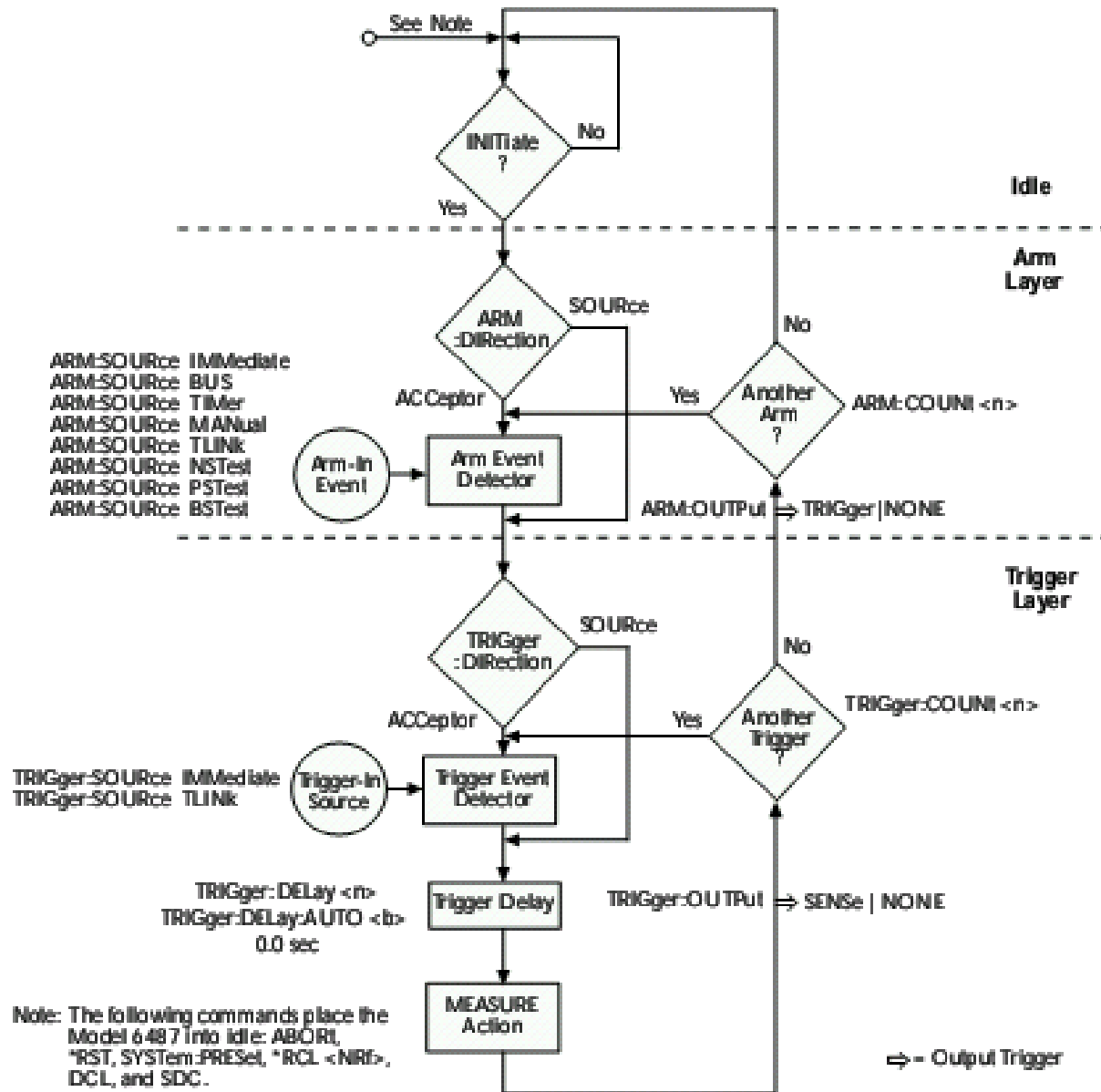
External triggering

Input and output triggers are received and sent via the rear panel TRIGGER LINK connector. The trigger link has six lines. At the factory line #2 is selected for output triggers and line #1 is selected for input triggers. These input/output assignments can be changed as previously explained in this section. The connector pinout is shown in [Figure 7-4](#).

Figure 7-4
Trigger link connection operation

Rear Panel Pinout	Pin Number	Description
	1	Trigger Link 1
	2	Trigger Link 2
	3	Trigger Link 3
	4	Trigger Link 4
	5	Trigger Link 5
	6	Trigger Link 6
Trigger Link	7	Ground
	8	Ground

Trigger model — remote operation



Trigger model operation

Once the instrument is taken out of idle, operation proceeds through the trigger model to perform a measurement (measure action).

NOTE *The following discussion focuses on the front panel trigger model (Figure 7-1). However, equivalent SCPI commands are included where appropriate.*

Event detectors and control sources

A control source holds up operation until the programmed event occurs and is detected. Note that there are two detector bypasses. A bypass around a detector is only enabled if the appropriate TLink control source is selected. See TLink control source (Arm-In and Trigger-In) as follows for details.

Arm-In source — The Arm-In control sources are explained as follows:

- **Immediate** (ARM:SOURce IMMediate) — Event detection for the arm layer is satisfied immediately allowing operation to continue into the trigger layer.
- **GPiB** (ARM:SOURce BUS) — Event detection for the arm layer is satisfied when a bus trigger (GET or *TRG) is received by the Model 6487.
- **Timer** (ARM:SOURce TiMer) — Event detection for the arm layer is immediately satisfied after the instrument leaves the idle state. Detection for each subsequent pass is satisfied when the programmed timer interval elapses. The timer resets to its initial state when the instrument goes back into idle.
- **Manual** (ARM:SOURce MAnual) — Event detection for the arm layer is satisfied by pressing the TRIG key. The Model 6487 must be in the local mode for it to respond to the TRIG key. Press LOCAL or send LOCAL 14 over the bus to place the Model 6487 in local.
- **TLink** (ARM:SOURce TLINk) — Event detection for the arm layer is satisfied when an input trigger via the TRIGGER LINK connector is received by the Model 6487. Note that if the source bypass is set to ONCE (ARM:DiRection SOURce), operation will initially loop around the source detector after the instrument leaves the idle state. Detection for each subsequent pass is satisfied by an input trigger. The bypass resets when the instrument goes into idle.
- **STest** (ARM:SOURce NSTest) — Event detection for the arm layer is satisfied when a negative-going pulse (via the SOT line of the Digital I/O) is received from a component handler (see [Limit Tests and Digital I/O in Section 8](#)).
- **STest** (ARM:SOURce PSTest) — Event detection for the arm layer is satisfied when a positive-going pulse (via the SOT line of the Digital I/O) is received from a component handler (see [Limit Tests and Digital I/O in Section 8](#)).
- **BSTest** (ARM:SOURce BSTest) — Event detection for the arm layer is satisfied when either a positive-going or a negative-going pulse (via the SOT line of the Digital I/O) is received from a component handler (see [Limit Tests and Digital I/O in Section 8](#)).

Trigger-In source — The Trigger-In control sources are explained as follow:

- **Immediate** (TRIGger:SOURce IMMediate) — Event detection for the trigger layer is satisfied immediately allowing operation to continue to perform a measurement.
- **TLink** (TRIGger:SOURce TLINk) — Event detection for the trigger layer is satisfied when an input trigger via the TRIGGER LINK connector is received by the Model 6487. Note that if the source bypass is set to ONCE (TRIGger:DiRection SOURce), operation will loop around the source detector on the initial pass through the arm layer. Detection for each subsequent pass is satisfied by an input trigger. The bypass resets when the Model 6487 leaves the trigger layer.

Rate

The RATE key selects the integration time of the A/D converter. This is the period of time the input signal is measured. The integration time affects the amount of reading noise, as well as the ultimate reading rate of the instrument. The integration time is specified in parameters based on a number of power line cycles (NPLCs), where 1 PLC for 60Hz is 16.67msec (1/60) and 1 PLC for 50Hz (and 400Hz) is 20msec (1/50).

In general, the Model 6487 has a parabola-like shape for its speed vs. noise characteristics and is shown in [Figure 4-1](#). The Model 6487 is optimized for the 1 PLC to 10 PLC reading rate. At these speeds (lowest noise region in the graph), The Model 6487 will make corrections for its own internal drift and still be fast enough to settle a step response <100ms.

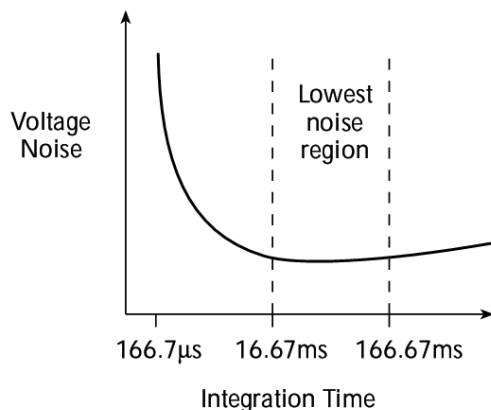
The rate setting is global for all ranges. Therefore, it does not matter what range is presently selected when you set rate.

There are two ways to set rate. You can select slow, medium, or fast by using the RATE key or you can set the number of power cycles from the NPLC menu that is accessed by pressing CONFIG / LOCAL (while in LOCAL) and then RATE.

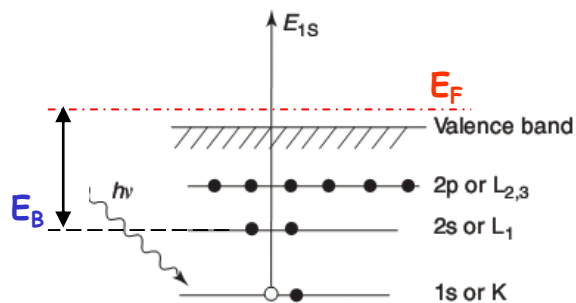
RATE Key — The RATE key selections are explained as follows:

- **SLOW** — Selects the slowest preset integration time (6 PLC for 60Hz or 5 PLC for 50Hz). The SLOW rate provides better noise performance at the expense of speed.
- **MED** — Selects the medium integration time (1 PLC). Select the MED rate when a compromise between noise performance and speed is acceptable.
- **FAST** — Selects the fastest preset integration time (0.1 PLC). Select the FAST rate if speed is of primary importance (at the expense of increased reading noise).

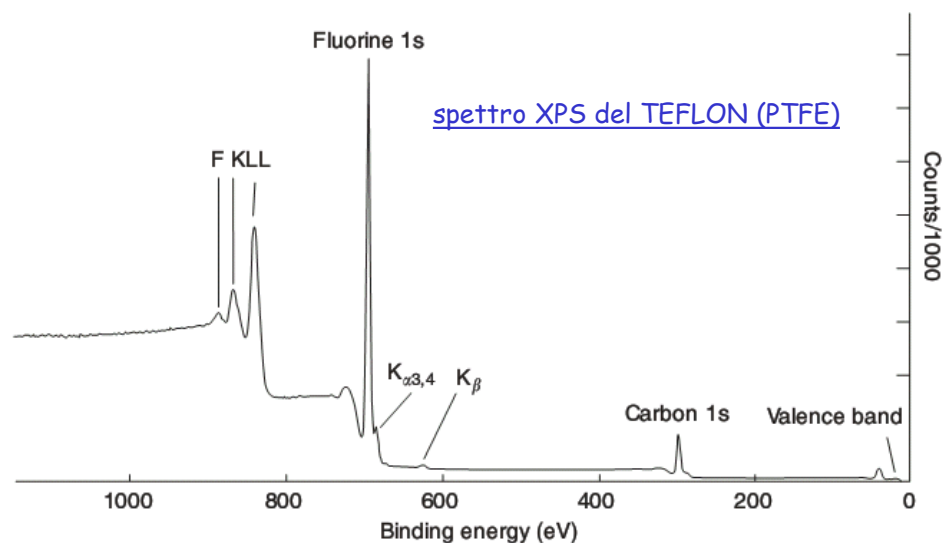
Speed vs. noise characteristics



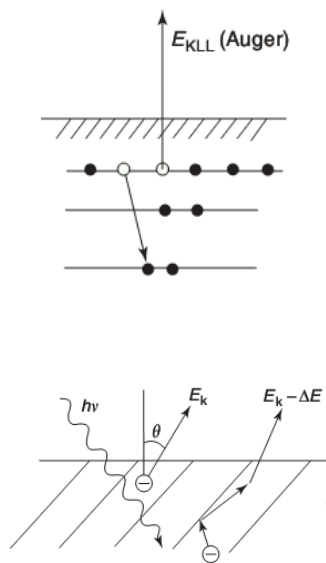
Spettroscopia di Fotoemissione



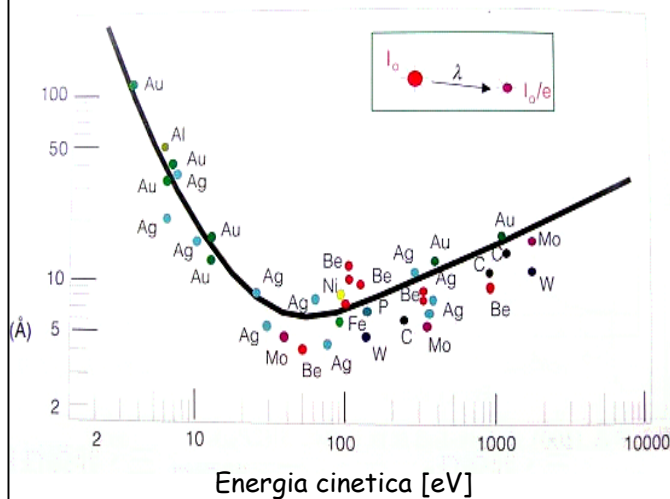
$$E_B = h\nu - E_K - \Phi$$



spettro XPS del TEFLON (PTFE)



cammino libero medio anelastico



EB dipende dallo stato di legame
(shift chimico)

